

Computer Vision-Based Analysis of Corn Leaf Orientation for Precision Agriculture

Group Internship in Collaboration with VeXtronics

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Abstract—This project presents a computer vision pipeline for analyzing corn leaf orientation and row alignment from standard top-down camera images. The work was carried out in collaboration with VeXtronics and aims to extract measurable plant-structure information that could support future robotic crop-monitoring and planting-quality evaluation. The pipeline combines YOLO11-pose detection with keypoint-based geometric analysis. A custom dataset of corn plants grown under controlled conditions was annotated with plant bounding boxes and three structural keypoints: the stem center and two visible leaf tips. The predicted keypoints were used to estimate plant position, fit crop-row directions, calculate positional and angular deviations, and compute alignment scores. In the final analysis, 146 plants were scored across Control and Experiment planting groups. The model achieved high corn detection performance, with an mAP50 of 0.993 for the corn class, while weed detection remained less reliable. The geometric results showed similar overall alignment scores for the Control and Experiment groups, with no statistically significant difference between them. These results show that the pipeline is a practical prototype for automated crop-structure analysis, while also highlighting limitations related to dataset size, image framing, weed confusion, and row reconstruction from close-up images.

Index Terms—Precision agriculture, corn leaf orientation, computer vision, YOLO11-pose, plant phenotyping, crop monitoring

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I Introduction

Corn is one of the most widely cultivated crops in the world, and improving how it is planted and monitored can support more efficient crop management. Besides factors such as soil quality, water availability, and planting density, the physical structure of the plant also plays an important role in its development. One structural characteristic that is especially relevant is leaf orientation. The direction and angle of corn leaves influence how sunlight is intercepted by the plant and how light is distributed within the canopy [1].

Leaf orientation is also relevant for planting quality. When leaves grow mainly along the crop row, neighboring plants may shade each other more strongly. In contrast, when leaves are more evenly distributed across the row, light can enter the canopy more effectively and self-shading may be reduced. Research on maize planting suggests that seed positioning can

influence early leaf orientation, meaning that the way a seed is placed in the soil may affect early plant architecture [2], [3].

Although leaf orientation is an important structural trait, measuring it manually is difficult. Field inspection is time-consuming, depends on the observer, and is not practical when many plants need to be evaluated. Manual measurements are also difficult to repeat consistently across different rows, growth stages, and field conditions. These limitations create a need for automated image-based methods that can measure structural plant traits in a more consistent and scalable way.

This need is especially relevant for agricultural robotics. In future corn-planting or crop-monitoring systems, visual feedback could be used to evaluate how plants developed after planting. For example, such a system could assess whether the intended seed placement and row structure resulted in the expected plant alignment and leaf distribution. This would allow planting quality to be evaluated not only by plant emergence, but also by the structural development of the crop.

This project was carried out as a group internship in collaboration with VeXtronics, a company working with agricultural technology and robotic systems. VeXtronics was interested in exploring whether computer vision could be used to evaluate corn plant structure from camera images and provide feedback for future robotic crop-monitoring or planting systems. Within this context, the objective of this project was to develop and evaluate a computer vision pipeline that detects individual corn plants, estimates structural keypoints, and converts these predictions into measurable indicators of leaf orientation and row alignment.

To achieve this, the pipeline combined YOLO11-pose detection with geometric analysis. The model predicted plant bounding boxes and structural keypoints, including the stem center and two leaf tips. These keypoints were then used to calculate plant position, row deviation, and leaf orientation. This makes it possible to move beyond simple plant detection and extract measurements that describe the structural arrangement of the crop.

A suitable dataset was an important part of this work. Publicly available corn image sources were used during early development to test and refine the detection pipeline. In addition, a custom dataset was created by growing corn plants under controlled conditions and collecting top-down images at different growth stages. These images were manually annotated using Roboflow so that the model could be trained and evaluated on plant bounding boxes and keypoint locations.

The final output of this project is an analysis pipeline that converts standard camera images into quantitative plant- and row-level measurements. The pipeline reports detected plant positions, fitted row directions, positional deviations from the row line, leaf-orientation deviations, and alignment scores. These outputs were used to compare rows and planting conditions, while also showing the limitations of the current setup, including sensitivity to image framing, dataset size, and imperfect row reconstruction from close-up images. As a result, the pipeline should be seen as a practical prototype for automated crop-structure analysis and a basis for future work on robotic planting-quality evaluation.

II Literature Review

II-A Precision Agriculture and Crop Monitoring

Precision agriculture uses data-driven technologies to monitor crops more accurately and support better field management

decisions. Instead of relying only on general field observations, technologies such as sensors, imaging systems, artificial intelligence, and automation allow crop conditions to be measured in more detail [4]. These technologies are important in modern agriculture because farmers aim to improve productivity while reducing unnecessary use of resources.

Computer vision has become an important part of precision agriculture because it allows crop information to be collected in a non-destructive and scalable way. Image-based systems have been used for plant detection, weed recognition, disease identification, fruit counting, and growth monitoring [5]. These applications show that camera-based analysis can reduce the need for manual inspection and make field monitoring more consistent. At the same time, many agricultural computer vision systems focus mainly on detecting or classifying visible objects, such as identifying whether a plant, weed, fruit, or disease symptom is present. For structural crop analysis, detection alone is often not sufficient because traits such as plant position, leaf direction, and row alignment require additional geometric interpretation after the plant has been located.

II-B Maize Leaf Orientation and Canopy Architecture

Maize canopy architecture strongly influences how efficiently plants capture and use sunlight. Leaf orientation is one of the traits that affects light distribution inside the canopy. When leaves overlap heavily or grow mainly along the crop row, shading between neighboring plants can increase. In contrast, a more favorable leaf arrangement can improve light penetration and support photosynthetic efficiency [1].

Previous research has shown that canopy structure is influenced by factors such as planting density, row spacing, hybrid type, and planting arrangement. Changes in plant density have been shown to affect canopy structure, light interception, and grain yield, indicating that maize productivity is related not only to the number of plants in a field but also to their spatial arrangement [1].

Seed orientation has also been investigated as a possible factor influencing maize leaf direction. Seed position in the soil has been shown to affect leaf orientation, while seed size and shape appear to have less influence [2]. Seed-oriented planting has also been reported to improve light interception, radiation use efficiency, and grain yield under certain experimental conditions [3]. Together, these findings suggest that seed placement can influence early maize architecture and may affect the resulting leaf distribution. The same studies also show a practical limitation: leaf orientation is often measured manually, visually assessed, or studied under controlled experimental conditions. Manual or highly controlled measurements are useful for agronomic research, but they are harder to apply repeatedly at field scale or integrate directly into automated agricultural systems.

II-C Plant Phenotyping and Structural Trait Measurement

Plant phenotyping focuses on measuring visible plant traits such as plant height, leaf area, leaf angle, canopy shape, and biomass. These traits provide information about plant development and help researchers evaluate the effects of genetics, environment, and management practices. Traditional phenotyping methods are often manual, which makes them slow and difficult to repeat consistently across many plants.

Automated phenotyping methods have been developed to improve the speed and reliability of plant trait measurement. Some systems use 3D imaging to capture detailed plant

architecture. For example, three-dimensional morphological features of corn leaves have been quantified to study leaf structure and orientation as important phenotypic traits [6]. These 3D methods provide detailed measurements, but they often require specialized imaging equipment or controlled scanning setups.

A practical gap therefore exists between high-precision phenotyping research and field deployment. While 3D systems can measure plant structure accurately, they may be less accessible for low-cost robotic or camera-based agricultural applications. Standard RGB camera imagery is easier to deploy, but RGB-based analysis also introduces limitations because depth information is not directly available and leaves may overlap in top-down images.

II-D Computer Vision and YOLO-Based Pose Estimation

Object detection is widely used in agricultural computer vision because many crop-monitoring tasks require specific plants or plant parts to be located in an image. YOLO, which stands for You Only Look Once, is a real-time object detection framework that predicts object locations and classes in a single processing stage [7]. Because of its balance between speed and accuracy, YOLO-based models are suitable for applications where images need to be processed efficiently.

Recent developments in YOLO-based models have expanded their use beyond object detection to tasks such as pose estimation. Pose estimation is relevant for structural crop analysis because it can predict specific keypoints rather than only object bounding boxes. YOLO11-pose, implemented through the Ultralytics framework, supports this type of keypoint-based prediction [8], [9].

Recent maize image datasets also show the growing interest in applying deep learning to crop monitoring. For example, the Maize Seedling Detection Dataset provides high-resolution RGB images for benchmarking object detection models under different maize field conditions [10]. These datasets are useful for evaluating detection performance, but detection alone does not provide all structural information needed for leaf-orientation analysis. Together, object detection, pose estimation, and maize image datasets show that computer vision can locate plants efficiently and extract structural image features. These predicted features still need to be converted into interpretable crop measurements, such as row deviation and leaf-angle deviation, before they can support planting-quality evaluation.

II-E Research Gap and Relevance of This Study

The literature reviewed above shows that maize leaf orientation is relevant for canopy structure, light interception, and crop development. Agronomic studies indicate that seed placement can influence early leaf direction, while phenotyping studies show that plant architecture can be measured in detail using manual or specialized imaging methods. At the same time, computer vision research demonstrates that crop plants can be detected from images and that pose-estimation models can predict structural keypoints.

The main gap is that these areas are usually treated separately. Agronomic studies explain why leaf orientation matters, but their measurements are often manual or performed under controlled experimental conditions. Phenotyping methods can provide detailed structural measurements, but they may require specialized equipment that is less suitable for practical robotic deployment. Detection-based computer vision methods can locate plants efficiently, but they often stop at object

localization and do not translate detections into crop-structure measurements.

This project addresses this gap by connecting plant detection, keypoint estimation, and geometric analysis in one practical pipeline. Instead of only detecting whether a corn plant is present, the proposed method predicts the stem center and leaf-tip keypoints and converts these points into interpretable measurements of plant position, row deviation, and leaf orientation. This makes it possible to evaluate the structure of corn plants from standard top-down RGB images.

For VeXtronics, this is relevant because the same type of measurement could support future agricultural robotic systems. In particular, the pipeline can provide feedback on whether plant growth and leaf orientation align with the intended row structure after planting. Such feedback could help assess planting quality, compare seed-orientation strategies, and support the development of more automated crop-monitoring systems.

III Methodology

The methodology of this project was structured as a sequential pipeline from image acquisition to plant-orientation analysis. The pipeline consists of five main stages: dataset collection, image annotation, model training, data augmentation, and geometric post-processing. The complete architecture of this end-to-end system is illustrated in Fig. III.1. Through these stages, top-down images of corn plants were converted into quantitative measurements of plant position, leaf orientation, and row alignment.

III-A Dataset Collection

The dataset for this project was created using two complementary sources. The primary dataset consisted of sweet corn plants cultivated under controlled greenhouse conditions in Heemskerk. The secondary dataset consisted of external corn image resources and related agricultural image sources, which were used during early model development to introduce additional variation in soil background, lighting, and plant spacing. The primary dataset was the main source for keypoint-based leaf-orientation analysis, while the secondary data mainly supported the plant detection stage.

a) Primary Data: Controlled Growth Setup

For the primary dataset, “suikermais” (sweet corn) was cultivated in a controlled greenhouse environment. Images were captured weekly to monitor plant development over time. The most relevant growth stages for this project were the early vegetative stages from V2 to V6, because the stem position and main leaf tips become increasingly visible during this period, as illustrated in Fig. III.2.

To keep the image acquisition process consistent, images were captured using a smartphone mounted on a fixed physical setup. This reduced variation in camera angle and distance between collection days and produced top-down images suitable for keypoint-based analysis. The primary dataset included both Control plants, planted without fixed seed orientation, and Experiment plants, planted with their long axis perpendicular to the crop row. Additional photographs of the data collection setup are provided in Appendix B.

b) Control and Experiment Groups

The primary dataset included two planting conditions. In the Control group, seeds were planted without a fixed orientation. In the Experiment group, seeds were manually placed with their long axis perpendicular to the crop row. These groups were later compared using the geometric alignment metrics.

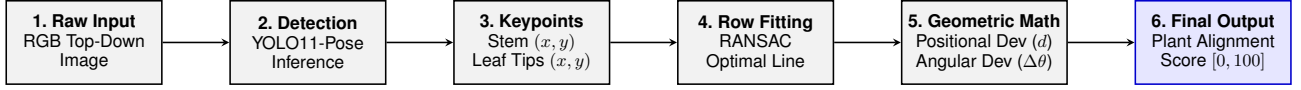


Fig. III.1: End-to-end system architecture. The pipeline processes raw agricultural imagery through a deep learning pose estimator to extract structural keypoints, which are then geometrically analyzed to output the crop alignment scores.

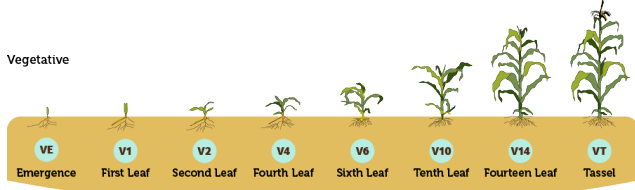


Fig. III.2: General vegetative growth stages of corn. The V2–V6 stages were used as a visual reference because the early leaf structure becomes increasingly visible during this period (image adapted from [11]).

c) Secondary Data: External Corn Spacing Dataset

In addition to the custom dataset collected for this project, an external corn image dataset was used during early model development. This dataset was obtained from the work of Brillhador et al. [12], who developed a computer vision method for measuring inter-plant spacing in corn fields. Their dataset includes corn field images from two conditions: conventional tillage and tillage after herbicide application. Because these images were originally collected for plant identification and spacing analysis, they were useful for testing and improving the corn detection stage of the pipeline.

The main reason for using this dataset was to add visual variation during early model development. Compared with the controlled greenhouse dataset, the external images contained different soil backgrounds, lighting conditions, plant spacing patterns, and field environments. This helped expose the model to more realistic agricultural conditions. However, the dataset did not include leaf-tip keypoint annotations. Therefore, it could not be used as ground truth for leaf-orientation estimation.

Some of the external images contained several corn plants in a single image. To use these images for single-plant training examples, a custom Python script was written to crop individual plant regions from the larger field images. This increased the number of single-plant examples available to the model. Overall, the secondary dataset mainly supported the plant detection stage, while the custom annotated dataset remained the main dataset for keypoint prediction and leaf-orientation analysis.

d) Weed Data

A supplementary weed dataset from Roboflow Universe was added to help the model distinguish corn plants from other vegetation [13]. This was important because real agricultural environments often contain weeds that can visually resemble early-stage corn plants, especially in top-down images. The weed images were used to introduce a separate weed class during training, allowing the model to learn examples of non-corn vegetation.

The purpose of this dataset was not to analyze weed species or evaluate weed detection as a separate research task. Instead, it was used to reduce false positive corn detections and to prevent weed plants from being included in the later geometric analysis. Since the dataset is an open-source Roboflow Universe dataset rather than a peer-reviewed agricultural dataset, it was used only as supplementary training data.

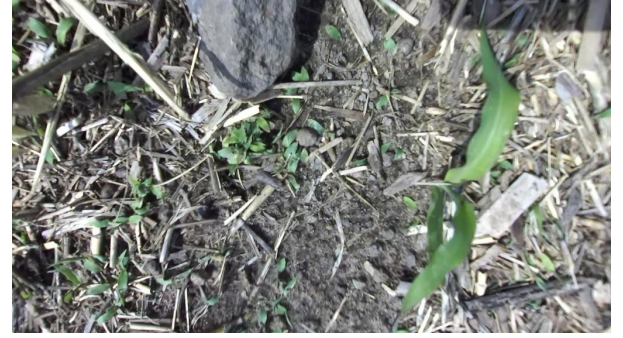


Fig. III.3: Example of a corn seedling surrounded by field weeds. The supplementary weed data was used to reduce false positive corn detections.

III-B Image Annotation

All collected images were manually annotated using Roboflow [14]. For each corn plant, a bounding box was created around the plant region. In addition, three structural keypoints were annotated: the stem center and two visible leaf tips. These annotations allowed the model to learn both plant localization and the keypoint positions required for orientation analysis.

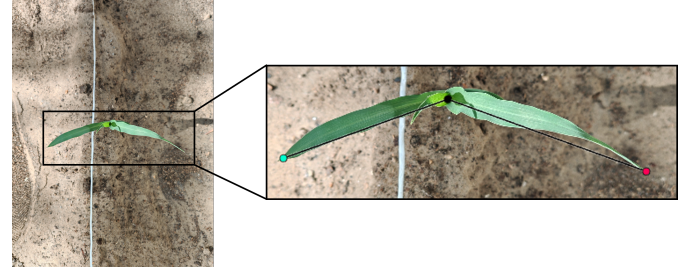


Fig. III.4: Manual keypoint annotation process. A raw greenhouse image is annotated with a bounding box and three structural keypoints. The stem is marked in black, while the two leaf tips are marked in cyan and red.

The annotation process required careful manual labeling, especially in images with overlapping leaves, shadows, weeds, or partially visible plant structures. Particular attention was given to the leaf-tip positions because these points directly determine the calculated leaf-orientation angle. Annotation quality was therefore important for both model training and the later geometric analysis. To improve consistency, the dataset was reviewed several times during labeling. After annotation, the dataset was divided into training, validation, and test subsets for model development and evaluation.

III-C Plant Detection and Pose Estimation

A YOLO pose estimation model, implemented through the Ultralytics framework, was used for corn plant detection and keypoint extraction [7], [8], [9]. YOLO was selected because it is a real-time object detection framework that provides a practical balance between speed and accuracy [7]. The pose estimation version was used because it can predict both a bounding box for plant localization and keypoints for structural analysis.

In this project, the model was trained to predict one bounding box around each corn plant and three keypoints: the stem

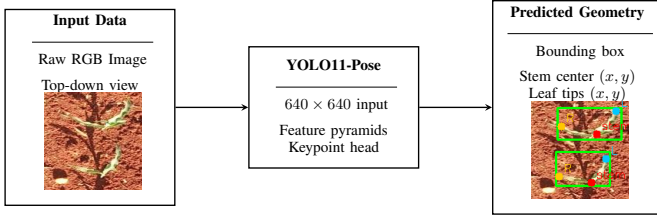


Fig. III.5: YOLO11-pose inference pipeline. The model receives a top-down RGB image and outputs a plant bounding box together with predicted stem and leaf-tip coordinates.

center and two leaf tips. These keypoints were required for the geometric analysis stage, where the stem center was used as the plant position and the line between the two leaf tips was used to estimate leaf orientation.

To compare accuracy and computational efficiency across model sizes and generations, three YOLO pose architectures were evaluated: YOLO11n-pose, YOLO11s-pose, and YOLO26s-pose. All models were initialized using pre-trained weights to improve convergence and reduce training time. The explicit Python initialization and evaluation scripts for these models are provided in Appendix C-B. During training, the models were trained for a maximum of 200 epochs using an input image resolution of 640×640 pixels. Early stopping was applied with a patience of 40 epochs to reduce the risk of overfitting.

Custom loss weights were used to place additional emphasis on keypoint prediction. This was important because small errors in leaf-tip localization can affect the calculated orientation angle. Therefore, the keypoint regression loss and keypoint objectness loss were increased during training so that the model prioritized the structural keypoints needed for the geometric analysis, rather than only optimizing bounding-box localization and class prediction.

Model performance was evaluated on the validation set using an Intersection over Union (IoU) threshold of 0.6. Bounding box localization and keypoint detection performance were measured using mean Average Precision (mAP) across different confidence thresholds.

The selected model output was not used as an end result by itself. Instead, the predicted bounding boxes and keypoints were passed to a geometric post-processing stage. This made it possible to convert image-level model predictions into plant-level and row-level measurements that could be compared between planting conditions.

III-D Data Augmentation

Data augmentation was applied during training to reduce overfitting and improve model robustness. The exact Python configuration dictionary used during model initialization is provided in Appendix C-A. This augmentation was especially important because agricultural images can vary due to lighting, shadows, camera position, soil background, and plant overlap.

Image augmentations were applied in the HSV color space to simulate different illumination conditions. Hue was varied by 2% to preserve the natural color of vegetation, while saturation and brightness were varied by 70% and 50%, respectively. These augmentations were used to improve robustness to direct sunlight, cloudy conditions, shadows, and local brightness differences in the crop images.

Geometric augmentations were also applied to simulate variations in camera height, centering, and alignment. These included random scaling of 40%, translation of 10%, rotation of 15° , shear of 3° , and minor perspective distortion. A horizontal flip probability of 50% was used. When horizontal

flipping was applied, the left and right leaf-tip keypoint labels were swapped to preserve annotation correctness. Vertical flipping was disabled because it would not match the expected orientation of the top-down row imagery.

Composite augmentations were also used to improve generalization. Mosaic augmentation was applied during training to combine four images into one composite image, increasing contextual variation. Random rectangle erasing was applied with a probability of 30% to simulate partial occlusion caused by overlapping leaves or weeds. Mixup augmentation was applied with a probability of 10% to provide additional regularization.

III-E Geometric Analysis

The geometric analysis stage converted each YOLO11-pose detection into a structured measurement of plant alignment. Because the model directly predicted the stem center and two leaf-tip keypoints, no separate segmentation or contour extraction step was required. The predicted keypoints provided the geometric primitives needed to estimate plant position and leaf orientation.

a) Plant Position and Leaf Axis

The stem keypoint was used as the plant position. The leaf axis was defined as the line passing through the two predicted leaf-tip keypoints. Since this axis has no inherent direction, its angle was normalized to the range $(-90^\circ, 90^\circ]$. If one or more keypoints had a confidence score below the selected threshold, the detection was not used for leaf-angle calculation. In such cases, the plant could still contribute to position-based row analysis, but not to orientation-based scoring.

b) Individual Leaf Angles

In addition to the overall leaf axis, individual leaf angles were calculated for each detected corn plant. While the leaf axis describes the undirected line passing through the two leaf-tip keypoints, the individual leaf angles describe the direction of each leaf relative to the stem.

Each individual leaf angle was calculated as a directional vector from the stem keypoint to a leaf-tip keypoint. For the left leaf, the angle was computed from the stem center to the left leaf tip. For the right leaf, the angle was computed from the stem center to the right leaf tip. Mathematically, these angles were obtained using the arctangent function:

$$\theta_{\text{left}} = \text{atan2}(y_{\text{left}} - y_{\text{stem}}, x_{\text{left}} - x_{\text{stem}}) \quad (\text{III.1})$$

$$\theta_{\text{right}} = \text{atan2}(y_{\text{right}} - y_{\text{stem}}, x_{\text{right}} - x_{\text{stem}}) \quad (\text{III.2})$$

The resulting angles were converted from radians to degrees for comparison with the row and leaf-axis angles.

Unlike the leaf-axis angle, individual leaf angles are directional. Therefore, opposite directions are not considered equivalent. For example, 0° and 180° describe the same undirected axis, but they represent opposite leaf growth directions. This distinction allows the pipeline to evaluate not only whether the two leaf tips form an axis perpendicular to the crop row, but also whether each individual leaf grows toward the expected side of the plant.

c) Individual Leaf-Angle Deviations

The expected individual leaf directions were derived from the expected leaf axis. One leaf was expected to grow along the perpendicular direction to the row, while the opposite leaf was expected to grow 180° away from it:

$$\theta_{\text{left,expected}} = \theta_{\text{expected}} \quad (\text{III.3})$$

$$\theta_{\text{right,expected}} = \theta_{\text{expected}} + 180^\circ \quad (\text{III.4})$$

Unlike the leaf-axis deviation, the individual leaf-angle deviations were calculated using directional angular differences. This means that opposite directions were not treated as equivalent:

$$\Delta\theta_{\text{left}} = |(\theta_{\text{left}} - \theta_{\text{left,expected}} + 180^\circ) \bmod 360^\circ - 180^\circ| \quad (\text{III.5})$$

$$\Delta\theta_{\text{right}} = |(\theta_{\text{right}} - \theta_{\text{right,expected}} + 180^\circ) \bmod 360^\circ - 180^\circ| \quad (\text{III.6})$$

These measurements were kept as additional structural descriptors because they can reveal asymmetric or folded leaf growth that may not be fully captured by the undirected leaf-axis angle alone.

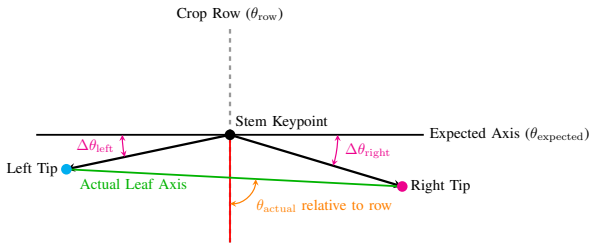


Fig. III.6: Directional deviation of individual leaves. The individual deviations $\Delta\theta_{\text{left}}$ and $\Delta\theta_{\text{right}}$ are calculated from the stem to each leaf tip and can describe asymmetric leaf growth that is not fully captured by the undirected leaf axis alone.

d) Optimal Row Line

For each crop row, an optimal reference line was defined to represent the expected row direction. When the planting layout was known, this line could be supplied manually using a point and an angle. Otherwise, the line was estimated from the detected stem centers using RANSAC, a robust model-fitting method designed to estimate model parameters in the presence of outliers [15].

RANSAC was selected instead of ordinary least squares because misaligned plants can bias a standard fitted line. By reducing the influence of such outliers, RANSAC can estimate a row line that better represents the majority of aligned stem centers. The row line was represented internally using a point and a unit direction vector, which avoids the slope singularity that occurs for near-vertical rows.

e) Per-Plant Deviations

Two deviations were calculated for each detected corn plant to quantify structural alignment, as shown in Fig. III.7. The positional deviation d was defined as the perpendicular pixel distance between the stem keypoint and the optimal row line.

To determine the angular deviation $\Delta\theta$, three angular components were calculated. First, the absolute angle of the optimal row line θ_{row} was established. Second, because seeds in the controlled trial were planted with their long axis perpendicular to the row direction, the expected leaf axis θ_{expected} was defined as the row direction rotated by 90° :

$$\theta_{\text{expected}} = \theta_{\text{row}} + 90^\circ \quad (\text{III.7})$$

Third, the actual leaf axis θ_{actual} was derived from the line passing through the two predicted leaf-tip keypoints. Since this leaf axis is an undirected line, opposite directions were treated as equivalent. The final axis-based angular deviation was therefore calculated as the smallest geometric difference between the actual leaf axis and the expected leaf axis:

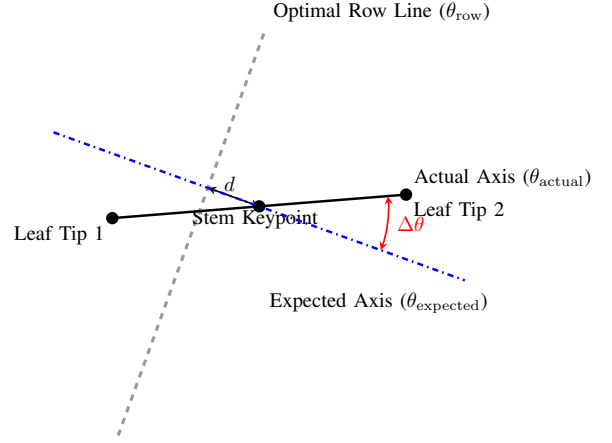


Fig. III.7: Geometric extraction of plant alignment. Positional deviation d is the perpendicular distance from the stem to the crop row. Angular deviation $\Delta\theta$ is the difference between the predicted leaf axis and the expected row-dependent leaf axis.

$$\Delta\theta_{\text{axis}} = |(\theta_{\text{actual}} - \theta_{\text{expected}} + 90^\circ) \bmod 180^\circ - 90^\circ| \quad (\text{III.8})$$

This axis-based angular deviation was used for the main alignment score, together with the positional deviation from the row line. Detections classified as weeds were removed before this stage so that they did not influence the row-line estimation or the final alignment score.

f) Alignment Scores

Each deviation was mapped to a sub-score in the range $[0, 100]$. The positional score (S_{pos}) decreased exponentially with increasing distance from the row line. This was used because small deviations from the row are acceptable, while large deviations indicate clearer spatial misalignment. It was calculated as:

$$S_{\text{pos}} = 100 \cdot e^{-d/\tau_{\text{pos}}}$$

where d is the perpendicular pixel distance from the stem to the optimal row line, and τ_{pos} is the positional tolerance constant.

The angular score (S_{ang}) was based on the axis-based angular deviation ($\Delta\theta_{\text{axis}}$) and decreased linearly with increasing angular error. A hard floor at zero was applied because a deviation of 90° represents a completely perpendicular, and therefore failed, alignment:

$$S_{\text{ang}} = 100 \cdot \max\left(0, 1 - \frac{\Delta\theta_{\text{axis}}}{\tau_{\text{ang}}}\right)$$

where τ_{ang} is the angular tolerance threshold in degrees. The tolerance constants (τ_{pos} and τ_{ang}) are tuning parameters that can be adjusted based on what constitutes an acceptable planting error.

The final per-plant alignment score (S_{plant}) was calculated by equally weighting the positional and angular sub-scores:

$$S_{\text{plant}} = 0.5 \cdot S_{\text{pos}} + 0.5 \cdot S_{\text{ang}}$$

Row-level scores were then obtained by averaging the plant scores within each row. In addition, the percentage of plants within the allowed positional tolerance was calculated to provide a directly interpretable measurement of planting alignment (e.g., the fraction of plants within a specific row that meet the required specification).

In a future autonomous monitoring or planting system, these standardized scores $[0, 100]$ could provide interpretable feedback on row quality. For example, consistently low angular scores could indicate that the seed orientation strategy or planting setup should be inspected further. However, such

closed-loop use would require validation on larger field datasets before deployment.

IV Results

IV-A Detection Results



Fig. IV.1: Example output of the geometric analysis pipeline on a test image. The model localizes the corn plant with a bounding box, predicts the stem and leaf-tip keypoints, and converts these points into leaf-direction vectors relative to the fitted crop row.

As illustrated in Fig. IV.1, the YOLO11-pose model was able to localize the corn plant and extract the structural keypoints required for the geometric analysis. The predicted bounding box, stem point, leaf-tip points, and fitted row line were visualized on the original image to make the output easier to interpret. This visualization shows how image-level model predictions were converted into plant-level measurements of position and leaf orientation.

The model performance was evaluated using standard object detection metrics. For the primary *corn* class, the model achieved an mAP50 of 0.993, indicating strong localization performance on the validation set. The secondary *weed* class achieved an mAP50 of 0.729, showing that weed detection was less reliable than corn detection. The validation confusion matrix in Fig. IV.2 shows that the main classification risk was confusion between background vegetation and corn. This is important because false positive corn detections could enter the geometric analysis if they are not filtered before row fitting and scoring.

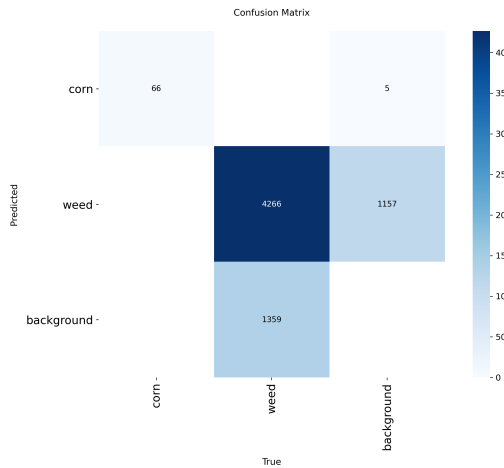


Fig. IV.2: Validation confusion matrix for the predicted classes: weed, corn, and background.

IV-B Leaf Orientation Results

Based on the geometric analysis, a total of 146 plants were scored to evaluate canopy consistency. The dataset was divided into two groups: 47 plants in the Control rows, representing the standard planting condition, and 99 plants in the Experiment rows, representing the oriented planting condition.

The metrics in Table IV.1 summarize the main geometric outputs of the pipeline. The mean alignment score is the final

combined score from 0 to 100, where a higher value indicates better combined positional and angular alignment. The mean positional deviation is the average perpendicular distance in pixels from the plant stem to the fitted row line. The mean angular deviation is the average difference in degrees between the predicted leaf axis and the expected leaf axis perpendicular to the row. The mean leaf deviation is based on the pooled individual leaf-angle deviations.

The overall mean alignment score across all scored plants was 28.3 out of 100. A comparison between the Control and Experiment groups is presented in Table IV.1.

The Control and Experiment groups had very similar mean angular deviations for the main leaf axis: 48.2° for the Control group and 48.0° for the Experiment group. This indicates that the main leaf-axis orientation did not differ meaningfully between the two planting conditions.

For the pooled individual leaf deviations, the Experiment group showed a lower mean deviation of 93.9° compared with 107.5° for the Control group. This suggests that the individual leaves in the Experiment group were slightly closer to the expected directional pattern. However, this improvement was not reflected in the overall alignment score because the Experiment group also showed worse positional alignment.

The Control group showed better spatial row alignment. Plants in the Control rows had a mean positional deviation of 41.7 pixels from the fitted row line, while plants in the Experiment rows had a higher mean positional deviation of 51.3 pixels. This indicates that the Experiment group showed greater spread around the expected row centerline.

Overall, the combined alignment scores were very similar: 27.8 for the Control group and 28.6 for the Experiment group. To test whether this difference was statistically meaningful, a Welch t-test was performed on the plant-level alignment scores. The test produced a p-value of 0.8493. Since $p > 0.05$, the difference between the two groups was not statistically significant. Therefore, the current dataset does not provide evidence that the oriented planting condition improved overall geometric alignment compared with the standard planting condition.

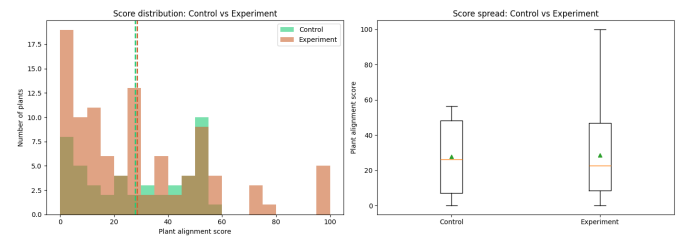


Fig. IV.3: Distribution of alignment scores for the Control and Experiment groups. The plot shows the spread of plant-level scores and supports the observation that the two groups have strongly overlapping score distributions.

IV-C Qualitative Robustness Observations

The pipeline was also inspected under image conditions that commonly affect agricultural computer vision systems, including lighting variation, shadows, background vegetation, and partial occlusion. These observations were qualitative and were used to identify strengths and limitations of the current prototype.

a) Lighting Variation

The model handled several lighting differences in the collected images, including brighter and darker regions caused by sunlight and shadows. This behavior is consistent with the use of HSV-based augmentation during training. However,

TABLE IV.1: Comparison of Alignment Metrics: Control vs. Experiment Groups

Metric	Control (Standard)	Experiment (Oriented)	Difference
Plants (n)	47	99	-
Mean Alignment Score	27.8	28.6	+0.8
Median Score	26.3	22.8	-3.5
Mean Positional Deviation	41.7 px	51.3 px	+9.6 px
Mean Angular Deviation	48.2°	48.0°	-0.2°
Mean Leaf Deviation (Pooled)	107.5°	93.9°	-13.6°

the current dataset is still limited, and the model should be evaluated on a larger set of field images before making strong claims about illumination robustness.

b) Background Vegetation and Noise

The corn class was detected reliably on the validation set, with an mAP50 of 0.993. However, the weaker weed performance, with an mAP50 of 0.729, indicates that background vegetation remains a limitation. In some cases, dense weed regions may be confused with crop structures. This could affect the geometric analysis if false positive detections are not removed before row-line fitting and alignment scoring.

c) Model Limitations

A main limitation of the current setup is the use of single-perspective, 2D top-down images. As corn plants grow into later vegetative stages, overlapping leaves can hide the stem center or leaf tips, making keypoint prediction more difficult. This can reduce the reliability of the calculated leaf axis and alignment score. In addition, although data augmentation helped reduce overfitting, the generalizability of the model to different soil types, growth stages, and field conditions still needs to be validated with a larger dataset.

IV-D Discussion

a) Interpretation of Results

The results show that the proposed pipeline can convert top-down RGB images into quantitative measurements of corn plant position, leaf orientation, and row alignment. The detection results indicate that corn localization worked well on the validation set, while weed detection remained less reliable. The geometric analysis made it possible to compare the Control and Experiment groups using the same positional and angular metrics.

From an agronomic perspective, the results do not support a strong conclusion that oriented seed placement improved overall canopy alignment in this dataset. Although the Experiment group showed a lower pooled individual leaf deviation, it also showed worse positional row alignment. As a result, the overall alignment scores were almost identical between the two groups, and the Welch t-test showed no statistically significant difference ($p = 0.8493$). This suggests that factors such as emergence quality, plant health, and row variability may have had a stronger effect than the intended seed orientation in this trial.

b) Architectural Choice: Pose Estimation vs. Semantic Segmentation

During the initial design phase, semantic segmentation was considered for leaf detection. However, a keypoint-based pose estimation approach was selected because the project required specific geometric points rather than full pixel-wise leaf masks. Semantic segmentation could provide more detailed leaf contours, but it would also require additional post-processing to extract a leaf axis from the segmented regions [16]. In contrast, the YOLO-pose approach directly predicted the stem center and leaf-tip coordinates needed for the angle calculations. This made the pipeline simpler and more suitable for the prototype stage.

c) Challenges

A major challenge during the project was the limited size and quality of the real dataset. Out of 250 planted seeds, 76 seedlings successfully emerged by the end of the growing period. Images were collected at multiple moments during development, resulting in 146 scored plant detections for the final geometric analysis. The limited emergence rate was affected by soil-borne corn rot, as documented in Appendix B. Because of the limited number of plants, the original physical rows had to be grouped into broader Control and Experiment categories for statistical comparison. This reduced the strength of the statistical analysis and may partly explain why no significant difference was found between the planting conditions.

d) Possible Improvements

Future work should expand the dataset across more plants, healthier growing conditions, and multiple growing seasons. This would make it possible to evaluate whether oriented planting has a measurable effect on leaf orientation under more reliable experimental conditions. From a technical perspective, the current 2D pose-estimation pipeline could also be extended with multi-view imaging or 3D reconstruction to reduce the effect of self-occlusion. Instance segmentation could also be explored in future work if complete leaf contours, leaf area, or canopy coverage are required in addition to the current keypoint-based measurements.

V Conclusion and Future Work

V-A Conclusion

This project developed a computer vision pipeline for evaluating corn leaf orientation and row alignment from standard top-down RGB images. The pipeline combined YOLO11-pose detection with geometric post-processing to localize individual corn plants, estimate structural keypoints, and convert these predictions into quantitative measurements of plant position, leaf orientation, and row alignment. The predicted stem center and leaf-tip keypoints made it possible to calculate positional deviation from the fitted row line and angular deviation from the expected leaf axis.

The results show that the proposed approach is technically feasible as a prototype for automated crop-structure analysis. The model achieved strong detection performance for the corn class, while weed detection remained less reliable. The geometric analysis allowed 146 plants to be scored and compared between the Control and Experiment planting groups. However, the overall alignment scores of the two groups were very similar, and the Welch t-test showed no statistically significant difference between them. Therefore, the current dataset does not provide evidence that the oriented planting condition improved overall geometric alignment compared with the standard planting condition.

The main contribution of this work is therefore not a final agronomic conclusion about seed orientation, but a working measurement framework that connects plant detection, keypoint estimation, and interpretable geometric analysis. This

framework can support future research on automated planting-quality evaluation, especially after validation on larger and more diverse datasets.

V-B Future Work

Several improvements are needed before the pipeline can be considered suitable for robust field deployment. First, the dataset should be expanded across more plants, healthier growing conditions, multiple growing seasons, and different field environments. A larger dataset would make it possible to evaluate the seed-orientation hypothesis more reliably and improve the generalizability of the model across different corn hybrids, soil backgrounds, lighting conditions, and growth stages.

Second, the current system relies on single-perspective 2D top-down images. This limits performance when leaves overlap or when the stem center and leaf tips are partially occluded. Future work could address this limitation by using multi-view imaging, low-flying drones, autonomous field rovers, or 3D reconstruction methods. These approaches could provide more complete information about plant structure and reduce the effect of self-occlusion in later vegetative stages.

Third, instance segmentation could be explored as an extension of the current keypoint-based method. While pose estimation is efficient for extracting stem and leaf-tip coordinates, segmentation could provide full leaf contours, allowing additional traits such as leaf area, canopy coverage, and plant shape to be measured. This would create a more complete phenotypic profile than geometric axes alone.

Finally, future versions of the pipeline should be optimized and tested for real-time use on edge devices. This would make the system more suitable for integration with agricultural robots or field-monitoring platforms. In the long term, combining early-stage structural measurements with later yield data could help evaluate whether plant alignment and leaf orientation are useful indicators for crop performance.

Code and Data Availability

The source code developed for this project, including the training pipeline, image processing scripts, live demonstration code, and geometric analysis implementation, is available in the project GitHub repository [17]. The repository contains the main Python scripts, notebooks, configuration files, and documentation required to run the pipeline. The README file explains the folder structure, required software environment, and commands for training the YOLO11-pose models, running validation, applying inference to new images, and reproducing the geometric leaf-orientation analysis. In addition, the notebooks and README file document the libraries used in the project, including the relevant package and version information needed to reproduce the setup.

The repository is not publicly released because the project was carried out in collaboration with VeXtronics and includes project-specific material. It is shared only for report assessment and grading purposes, in agreement with VeXtronics. The custom annotated dataset is also not publicly released at this stage because it was collected as part of the group internship and may contain project-specific data. However, the annotation format, preprocessing steps, training setup, and evaluation procedure are documented so that the pipeline can be reproduced with equivalent YOLO-pose annotated image data.

The external corn spacing dataset used during early model development is publicly available from the original authors

[12]. Additional implementation details, including selected training and evaluation snippets, are provided in Appendix C.

Technology Statement

OpenAI's ChatGPT was used during the preparation of this report to support line-by-line code debugging, grammar checking, and sentence-level clarity improvements [18]. The tool was used only as an assistant. The experimental work, implementation decisions, results, analysis, and final interpretation remained the responsibility of the authors. All AI-assisted suggestions were reviewed, verified, and edited by the team before inclusion in the final report.

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Appendix A

Individual Contributions

This appendix summarizes the individual contributions of each team member during the group internship project. The work was divided across dataset preparation, model development, annotation, testing, implementation, demonstration, research, and report writing. Although each team member focused on different parts of the project, the final system was developed collaboratively and the main project decisions were discussed as a team.

A-A Merve Şafak

Merve contributed to the preparation and collection of external online dataset material, including corn image resources used during early model development. She contributed to research on precision agriculture, corn leaf orientation, and related literature. She also supported the live demonstration code, helped improve the readability and structure of the general pipeline, and contributed to report writing and editing.

A-B Athanasios Poulis

Athanasios contributed to the main computer vision pipeline, including YOLO11-pose model training, validation, and evaluation. His work involved testing the pose-estimation approach, inspecting model outputs, comparing detection and keypoint results, and supporting the final detection and keypoint-estimation workflow. He also contributed to the live demonstration and report writing.

A-C Miquel Suasso de Lima de Prado

Miquel contributed to model testing, dataset annotation, and the evaluation of different model variants. His work supported the comparison of training configurations and helped improve the reliability of the detection and keypoint-prediction process. He also contributed to pipeline testing, checking model outputs, result interpretation, and report writing.

A-D Alecsandru Kreefft-Libiu

Alecsandru contributed to the collection of the real project dataset, including planting the corn, monitoring plant growth, taking images, and supporting the creation of the controlled image dataset. He also contributed to data annotation, testing the pipeline on collected images, the live demonstration, and report writing.

A-E Collaborative Work

All team members contributed to project discussions, testing sessions, debugging, final integration, and preparation of the final report. The main project decisions, including the pipeline design, interpretation of results, and preparation of the final presentation, were made collaboratively. Overall, the workload was distributed across the team, with each member contributing to both technical development and project documentation.

Appendix B

Physical Data Collection Setup

The physical data collection setup was designed to capture consistent top-down images of the corn plants across different collection days. A smartphone mounting rig was used to keep the camera angle and distance as stable as possible during image acquisition. This helped reduce unwanted variation caused by changes in camera position and made the collected images more suitable for keypoint-based leaf-orientation analysis.

Figure B.1 shows the physical setup used during image collection.



Fig. B.1: This is similar to the rig that was used to capture the top-down images during data collection.

B-A Field Preparation and Seed Placement

To establish a ground truth for the optimal row lines, the greenhouse soil was prepared and measured prior to cultivation. String lines were placed across the planting area to enforce strict linear rows and in order to keep track of the seeding line. During the planting phase, a measuring tape was used to have consistent spacing. Most seeds were manually placed in the soil with its long axis oriented strictly perpendicular to the string line. This controlled placement was necessary to establish the theoretical expected leaf growth axis used later in the geometric analysis pipeline. The other seeds were used as a control group, where we randomly planted the seeds.



Fig. B.2: Greenhouse preparation using string lines to enforce straight row trajectories.



Fig. B.3: Controlled seed placement perpendicular to the row to establish expected growth axes.

B-B Plant Conditions and Canopy Development

Throughout the data collection period, the structural development of the canopy was documented. The top-down images capture the variance in leaf droop, asymmetry, and self-occlusion as the plants grew. These images highlight the visual complexity that the YOLO-pose architecture had to successfully navigate to extract consistent structural keypoints.



Fig. B.4: Top-down perspectives of varying plant conditions, demonstrating natural asymmetric leaf droop, soil background variations, and differing vegetative growth stages.

B-C Agricultural Challenges: Soil-Borne Pathogens

As noted in the discussion regarding dataset limitations, the physical yield of the controlled greenhouse trial was heavily impacted by real-world agricultural variables. Figure B.5 illustrates an excavated seed suffering from soil-borne corn rot. This pathogen was primarily responsible for the reduction in the seedling emergence rate, which resulted in the restructuring of the test rows during the final analysis.



Fig. B.5: An excavated seed exhibiting soil-borne corn rot, which impacted the final emergence rate of the physical trial.

Appendix C Pipeline Code Snippets

This appendix provides selected code snippets from the training pipeline to support reproducibility. The full implementation is available in the project GitHub repository cited in the Code and Data Availability section.

C-A YOLO Augmentation Configuration

The augmentation parameters used during YOLO11-pose training are shown in Listing 1. These settings were chosen to increase robustness to common image variations, including lighting changes, scale differences, camera misalignment, and partial occlusion.

```
# Augmentation parameters passed to YOLO's trainer.
AUG = dict(
    # Color jitter: simulates different illumination
    # conditions
    hsv_h=0.02,      # hue shift
    hsv_s=0.70,      # saturation variation
    hsv_v=0.50,      # brightness variation

    # Geometric jitter: simulates small camera-
    # position changes
    degrees=15,      # rotation
    translate=0.10,  # translation
    scale=0.40,      # scaling
    shear=3,         # shear
    perspective=0.0005, # perspective distortion

    # Flipping
    flipplr=0.5,
    flipud=0.0,

    # Composite and occlusion-based augmentation
    mosaic=1.0,
    mixup=0.10,
    erasing=0.30,
)
```

Listing 1: Custom data augmentation dictionary applied during model training.

C-B YOLO Training and Evaluation Initialization

The following Python scripts demonstrate the initialization of the pose estimation models (nano and small) and the extraction of the geometric mean Average Precision (mAP) metrics.

```
from ultralytics import YOLO

# Load pose models with pre-trained weights
model_n = YOLO('yolo11n-pose.pt')
model_s = YOLO('yolo11s-pose.pt')
model_26s = YOLO('yolo26s-pose.pt')

# Standardized training arguments (applied to all
# models)
train_args = dict(
    data=str(data_yaml_path),
    epochs=200,
    imgsz=640,
    batch=8,
    workers=2,
    patience=40,
    device=0,
    project=str(PROJECT_ROOT / 'runs'),
    exist_ok=False,
    pose=30.0,      # Custom keypoint
    regression_loss_weight,  # Custom keypoint
    kobj=2.0,      # Custom keypoint
    objectness_loss_weight,  # Custom keypoint
    **AUG,          # Unpack
    augmentation_settings (from Appendix C-A)
)

# Kick off training (Example shown for the 's'
# architecture)
results_s = model_s.train(name='yolo11s-pose_corn',
    **train_args)
```

Listing 2: Model initialization and training loop for the YOLO pose architectures.

```
RUNS_DIR = PROJECT_ROOT / 'runs'
BEST_S = RUNS_DIR / 'yolo11s-pose_corn' / 'weights'
/ 'best.pt'

print(f"Evaluating:_{BEST_S}")
model = YOLO(str(BEST_S))

# Run one pass over the validation set
# conf=0.001 accepts all predictions to compute
# precision/recall across the whole curve
# iou=0.6 is the threshold for non-max suppression
metrics = model.val(data=str(data_yaml_path), imgsz
    =640, conf=0.001, iou=0.6)

# Extract and print overall metrics
print(f"\nBox_mAP50:_{metrics.box.map50:.3f}")
print(f"Box_mAP50-95:_{metrics.box.map:.3f}")
print(f"Pose_mAP50:_{metrics.pose.map50:.3f}")
print(f"Pose_mAP50-95:_{metrics.pose.map:.3f}")

# Extract Precision and Recall
mp, mr, m_ap50, m_ap = metrics.box.mean_results()
print(f"\nBox_Precision:_{mp:.3f}|_Box_Recall:_{mr:.3f}")

pp, pr, p_ap50, p_ap = metrics.pose.mean_results()
print(f"Pose_Precision:_{pp:.3f}|_Pose_Recall:_{pr:.3f}")

# Per-class breakdown
print("\nPer-class_detection_metrics:")
for class_idx, class_name in model.names.items():
    p, r, ap50, ap = metrics.box.class_result(
        class_idx)
    print(f"_{class_name:6s}:_P={p:.3f}_R={r:.3f}_
        _mAP50={ap50:.3f}")
```

Listing 3: Validation and metric extraction script for bounding box and keypoint accuracy.